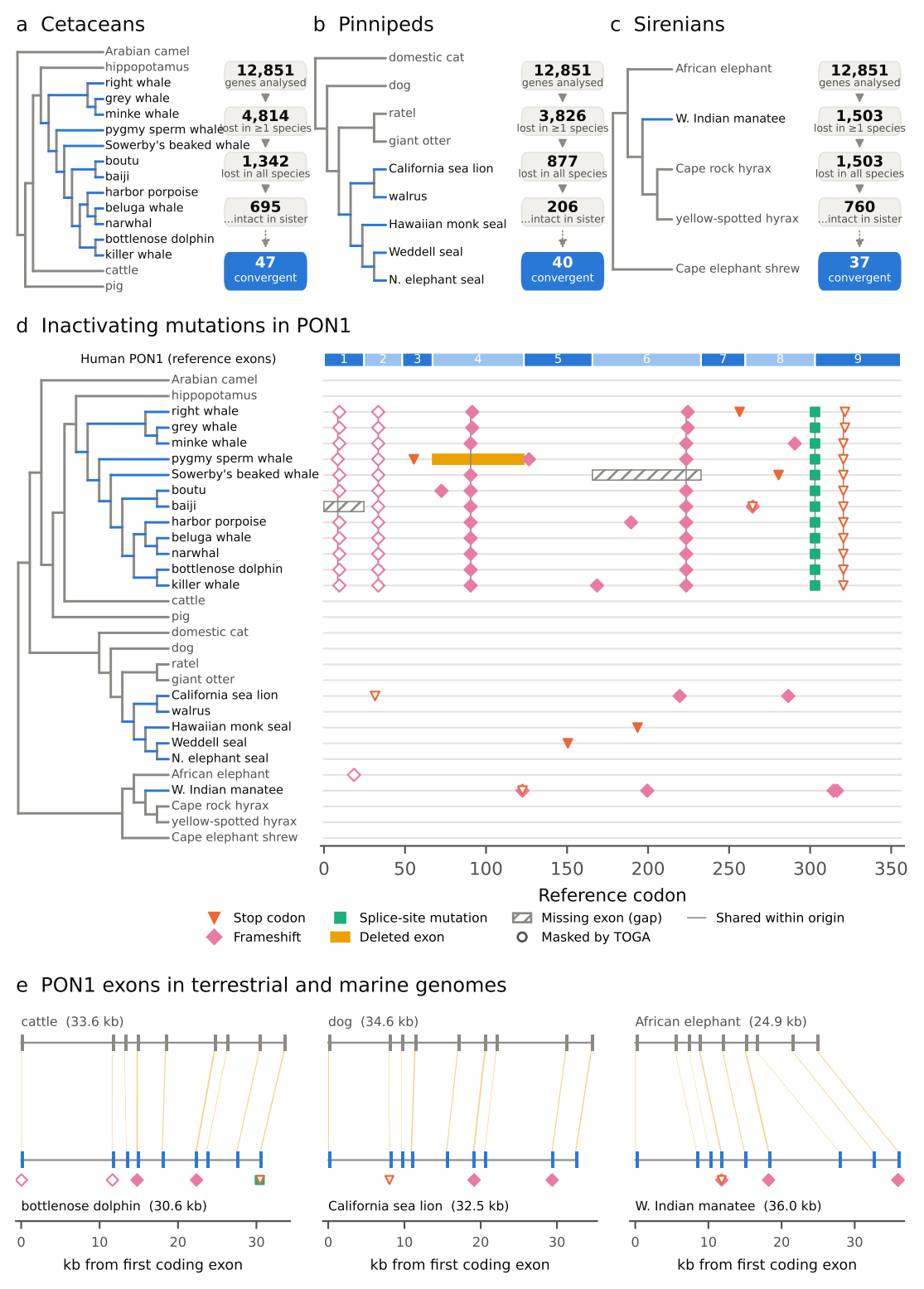
a Cetaceans
Arabian camel
hippopotamus
right whale
grey whale
minke whale
pygmy sperm whale
Sowerby's beaked whale
boutu
baiji
harbor porpoise
beluga whale
narwhal
bottlenose dolphin
killer whale
cattle
pig
12,851 genes analysed
4,814 lost in ≥1 species
1,342 lost in all species
695 ...intact in sister
47 convergent
b Pinnipeds
domestic cat
dog
ratel
giant otter
California sea lion
walrus
Hawaiian monk seal
Weddell seal
N. elephant seal
12,851 genes analysed
3,826 lost in ≥1 species
877 lost in all species
206 ...intact in sister
40 convergent
c Sirenians
African elephant
W. Indian manatee
Cape rock hyrax
yellow-spotted hyrax
Cape elephant shrew
12,851 genes analysed
1,503 lost in ≥1 species
1,503 lost in all species
760 ...intact in sister
37 convergent
d Inactivating mutations in PON1
Human PON1 (reference exons)
1
2
3
4
5
6
7
8
9
0
50
100
150
200
250
300
350
Reference codon
Stop codon
Frameshift
Splice-site mutation
Deleted exon
Missing exon (gap)
Masked by TOGA
Shared within origin
e PON1 exons in terrestrial and marine genomes
cattle (33.6 kb)
dog (34.6 kb)
African elephant (24.9 kb)
bottlenose dolphin (30.6 kb)
California sea lion (32.5 kb)
W. Indian manatee (36.0 kb)
0
10
20
30
kb from first coding exon